

# Year 8 Test

Term 3 2017

## Equations, Statistics and Time

### Resources Allowed

- 1 A4 page of notes
- Calculator

Name: \_\_\_\_\_

Score: \_\_\_\_/55

### 1. [17 marks]

For each of the following equations find the value of  $x$ .

a)  $x + 4 = 12$

$$x = 12 - 4$$

$$x = 8$$

b)  $b - 15 = 62$

$$b = 62 + 15$$

$$b = 77$$

c)  $3x + 2 = -13$

$$3x = -13 - 2$$

$$x = -15 \div 3$$

$$x = -5$$

d)  $3d = 63$

$$d = 63 \div 3$$

$$d = 21$$

e)  $5e - 4 = 21$

$$5e = 21 + 4$$

$$e = 25 \div 5$$

$$e = 5$$

h)  $3(x + 12) = 18$

$$x + 12 = 18 \div 3$$

$$x = 6 - 12$$

$$x = -6$$

f)  $\frac{2k-2}{5} = 6$

$$2k - 2 = 6 \times 5$$

$$2k = 30 + 2$$

$$k = 32 \div 2$$

$$k = 16$$

i)  $\frac{x}{4} - 6 = -1$

$$\frac{x}{4} = -1 + 6$$

$$x = 5 \times 4$$

$$x = 20$$

g)  $3x = x + 4$

$$3x - x = 4$$

$$2x = 4$$

$$x = 2$$

j)  $4(l + 1) = 2(l - 3)$

$$4l + 4 = 2l - 6$$

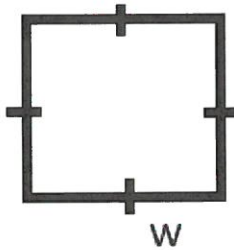
$$4l - 2l = -6 - 4$$

$$2l = -10$$

$$l = -5$$

2. [3 marks]

A square has a perimeter of 36cm.



- a) Write an equation using  $w$  as the width,

$$4w = 36$$

- b) Using your equation from a), determine the width of the square.

$$w = 36 \div 4$$

$$w = 9$$

The width is 9cm

3. [3 marks]

Rhonda's mum is 3 times older than her and her dad is 2 years older than her mum. If her dad is 41 then

- a) Set up an equation to find Rhonda's ages ( $r$ ).

$$3r + 2 = 41$$

- b) Find Rhonda's age.

$$3r + 2 = 41$$

$$3r = 41 - 2$$

$$r = 39 \div 3$$

$$r = 13$$

Rhonda is 13 yrs old

4. [3 marks]

A number is tripled, and then 2 is added. This gives the same result as if the number were quadrupled. Setup and solve an equation to find the original number.

$$3x + 2 = 4x$$

$$2 = 4x - 3x$$

$$2 = x$$

Original number is 2

5. [10 marks]

Consider the following set of data

73, 57, 73, 62, 20, 69, 62, 73, 65, 67

a) Draw a stem and leaf plot for this data.

| stem | leaf      |
|------|-----------|
| 2    | 0         |
| 3    |           |
| 4    |           |
| 5    | 7         |
| 6    | 2 2 5 7 9 |
| 7    | 3 3 3     |

b) Calculate;

I. Mean

$$621 \div 10 = 62.1$$

II. Median

$$(65 + 67) \div 2 = 66$$

III. Mode

73

IV. Range

$$73 - 20 = 53$$

b) Which score in the set of data is an outlier?

20

c) If the outlier is removed, what effect will this have on the;

Mean: Increases (66.8) and Mode: No change (73)

6. [3 marks]

What type of data are the following?

Choose from categorical, discrete numerical or continuous numerical:

a the number of cans of Coke in a Coke machine

Discrete Num

b the favourite ice cream of each of the students in a class

Categorical

c the length of time that a year 8 student can spend running

Continuous Num

7. [3 marks]

Leroy surveys his entire Year 8 class to find out the best holiday destination in Australia?

a) Is this a sample or a census? Why?

Sample because we are only asking for part not the whole of a population

b) How is this survey biased?

• All the same age

OR • All live in the same area etc

8. [3 marks] Complete the table

| 24 hour time | 12 hour time |
|--------------|--------------|
| 0017hrs      | 12:17am      |
| 1440 hrs     | 2:40pm       |
| 1959hrs      | 7:59pm       |

9. [4 marks] Convert the following

- a) 180 sec = 3 min ✓
- b) 3 years = 1095 days ✓
- c) 4 weeks =  $4 \times 7 \times 24 = 672$  hours //

10. [2 marks]

Jarod went to the movies with some friends. If the movie started at 6:45pm and the movie went for 1 hour and 30 mins, at time did the movie finish?

- d) 12 hour time 8:15pm ✓
- e) 24 hour time 2015hrs ✓

11. [2 marks]

A taxi departs at 18:52 from Sydney and arrives at 21:33 at Melbourne. How long did the taxi take for the trip? *In the same timezone\**

$$\begin{aligned}
 1852 &\rightarrow 2052 = 2\text{hrs} \\
 2052 &\rightarrow 2100 = 8\text{mins} \\
 2100 &\rightarrow 2133 = 33\text{mins} \\
 &\quad \underline{2\text{hrs } 41\text{mins}} \quad \checkmark
 \end{aligned}$$

12. [2 marks]

Sarah in Sydney is expecting a phone call from Fiona in Manilla. Manilla is 2 hours behind Sydney time.

- a) If Sarah calls at 10:45am, what time is it for Fiona?

8:45am ✓

- b) If Fiona calls at 3:30pm, what time is it for Sarah?

5:30pm ✓